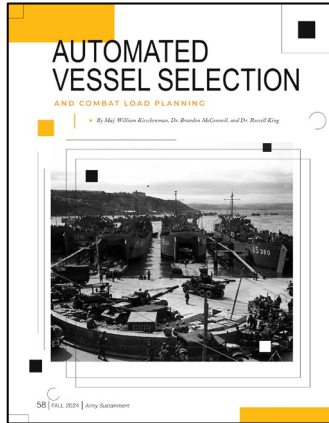




The 2-D Bin Packing Problem with Multiple Levels of Prioritization: A Spatial Optimization Perspective



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<https://go.ncsu.edu/priority-pack>



Purpose: Introduce a novel 2-D Bin Packing (BPP) framework integrating multi-level spatial prioritization and demonstrate its effectiveness for complex operational planning using a scalable decomposition method and novel lower bounds to quickly generate and assess high quality solutions.

Agenda:

- Introduction & Motivation.
- Literature Landscape.
- Prioritized 2D-BPP Framework.
- Solution Techniques & Lower Bounds.
- Key Computational Findings.
- Conclusion & Future Directions.

Introduction

Motivation

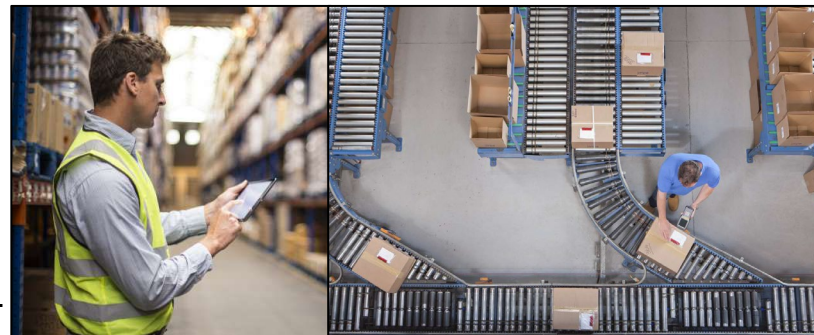
- **Classical 2-D Bin Packing Problems (2D-BPP):**
 - primarily focus on spatial efficiency (minimizing bins/waste).
- **Real-World Discrepancy:**
 - Many scenarios require more than just efficient fitting.
 - Layouts optimized *only* for space can be operationally inefficient or infeasible.
- **Critical Gap: Standard methods often overlook operational priorities:**
 - Placement sequence (e.g., for unloading).
 - Keeping related items together (group cohesion).
 - Proximity to access points (e.g., doors, ramps).
- **Our Contribution:** A nuanced approach that directly incorporates spatial prioritization into the packing optimization.

Research Question: *How can we incorporate multiple levels of prioritization into the two-dimensional bin packing problem to better capture real-world operational needs?*

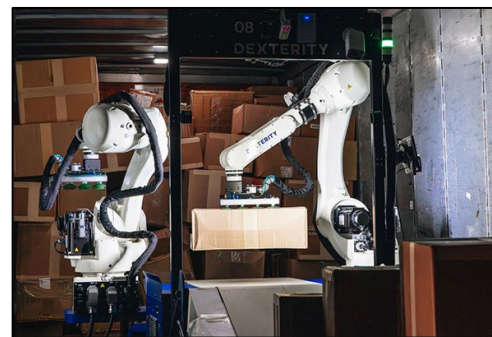
Introduction

Applications

- **Warehouse Operations / E-Commerce:**
 - Grouping items for customer orders.
 - Sequencing for efficient picking/packing.
 - Placing high-frequency items near dispatch areas.
- **Logistics & Transportation:**
 - Truck loading for multi-stop deliveries (LIFO/FIFO principles).
 - Optimizing cargo placement for faster handling.
- **Military Combat Loading:**
 - Sequencing equipment for tactical off-load (“order of march”).
 - Maintaining unit integrity (group cohesion).
 - Prioritizing critical assets near exit points.
 - *Relevance:* Joint Publication 3-02, historical amphibious landing lessons.



UPS continuously assesses warehouse operations and to propose processing improvements and give equipment handling and storage recommendations (UPS, 2023).



Dexterity AI and FedEx are developing two-armed robotic truck loading (Dexterity AI, 2023).

Literature Review

Bin Packing Literature (BPP):

- Focus: Efficiently packing items, often *minimizing bins/waste*; NP-hard (Bortfeldt and Wäscher (2013), Garey (1979)).
- 2D-BPP methods (MILP, heuristics) *traditionally optimize space, not operational priorities* (Lodi et al. (2002), Castro et al. (2011)).
- We adapt *core geometric constraints* (Paquay et al. (2016), Fontaine & Minner (2023)) but introduce a *novel prioritization-driven objective*.

Facility Layout Planning (FLP):

- Focus: Arranging departments to minimize weighted distances/costs (Meller et al., 1998).
- Models like Quadratic Assignment Problem (QAP) or Unequal Area FLP (UA-FLP) share geometric aspects but differ in objective (Koopmans & Beckman (1957), Bozer & Meller (1997)).
- **Gap:** FLP rarely incorporates flexible, multi-tiered item/group specific priorities needed for operational packing.

Spatial Optimization in Packing:

- Some research considers accessibility, sequencing, or load balancing (e.g., Bischoff (1995) on multi-drop container loading).
- Container Loading (CLP) literature explores unloading sequences (Altarazi, 2013), but explicit distance-based multi-level prioritization is rare (e.g., Bortfeldt and Wäscher (2013) survey).
- **Our Work:** Unifies BPP rigor with FLP-style distance objectives via a flexible prioritization matrix, addressing a critical gap.

A Prioritized 2D-BPP Framework

Sets

Items:	$I = \{1, 2, \dots, n\}$	Set of n rectangular items
Bin Dimensions:	$C = \{1, 2\}$	Corresponds to (x, y) dimensions of the bin
Item Dimensions:	$D = \{1, 2\}$	Corresponds to (x, y) dimensions of an item

Data and Parameters

Bin Length:	$L > 0$	Fixed length of the single rectangular bin
Bin Width:	$W > 0$	Fixed width of the single rectangular bin
Item i Original Length:	$p_i \geq 0$	$\forall i \in I$
Item i Original Width:	$q_i \geq 0$	$\forall i \in I$
Prioritization Matrix:	π_{ik}	$\forall i, k \in I, i \leq k$ π_{ii} : Weight for item i to access point $\pi_{ik}, i < k$: Weight for items i, k cohesion
Bin Access Point Coords:	(x^o, y^o)	Known coordinates (e.g., $x^o = 0, y^o = W/2$ for center-left)

Decision Variables

Item i Left x-coord:	$x_i \geq 0$	$\forall i \in I$
Item i Bottom y-coord:	$y_i \geq 0$	$\forall i \in I$
Item i Right x-coord:	$x_i^r \geq 0$	$\forall i \in I$
Item i Upper y-coord:	$y_i^u \geq 0$	$\forall i \in I$
Item i left of item k (x-dim):	$a_{ik}^x \in \{0, 1\}$	$\forall i, k \in I, i \neq k$
Item i bottom of item k (y-dim):	$a_{ik}^y \in \{0, 1\}$	$\forall i, k \in I, i \neq k$
Orientation of item i :	$t_{i,c,d} \in \{0, 1\}$	$\forall i \in I, c \in C, d \in D$
Rectilinear dist. (x) for pair (i, k) :	$\rho_{ik}^x \geq 0$	$\forall i, k \in I, i \leq k$
Rectilinear dist. (y) for pair (i, k) :	$\rho_{ik}^y \geq 0$	$\forall i, k \in I, i \leq k$

Goal: To develop and evaluate a novel mathematical framework for the 2D Bin Packing Problem that integrates multiple levels of spatial prioritization directly into the optimization process.

Key Idea: Integrate FLP Concepts into BPP.

- Adapt distance-minimization objectives from Facility Layout Planning (FLP).
- Use a **Prioritization Matrix (π)** to encode spatial preferences:
 - Diagonal ($\pi_{i,k}$ where $i = k$):** Priority weight for placing item i close to the bin access point (x^o, y^o) . *Higher value = stronger pull to access point.*
 - Off-Diagonal ($\pi_{i,k}$ where $i < k$):** Priority weight for placing items i and k (often same group) close to each other (cohesion). *Higher value = stronger pull together.*
- Optimize item placement by minimizing a **Weighted-Distance Objective Function** (rectilinear distance):

$$\min \sum_{i \in I} \sum_{k \in I, i \leq k} \pi_{ik} (\rho_{ik}^x + \rho_{ik}^y)$$

- This simultaneously handles proximity to access points and group cohesion, balanced by the weights in $\pi_{i,k}$.

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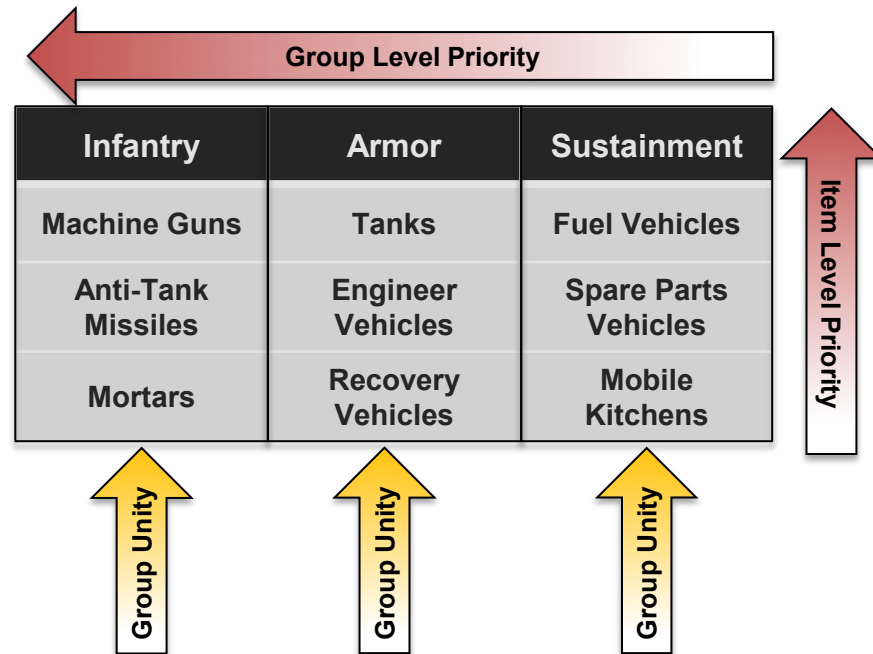
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Prioritized 2-D BPP Formulation: Constraints

Item Sizing & Orientation:

- Ensures item dimensions are respected based on chosen orientation (orthogonal packing).

Non-Overlapping:

- Standard BPP constraints preventing items from occupying the same space.
- Binary variables determining relative positioning.

Distance Calculation

- Linearize rectilinear distances between item centroids and item centroids to access point for the objective function.

Boundary & Domain Constraints

- Items within bin limits; variable types defined.

$$x_i^r - x_i = t_{i,1,1}p_i + t_{i,1,2}q_i \quad \forall i \in I \quad (2)$$

$$y_i^r - y_i = t_{i,2,1}p_i + t_{i,2,2}q_i \quad \forall i \in I \quad (3)$$

$$a_{ik}^x + a_{ki}^x + a_{ik}^y + a_{ki}^y \geq 1 \quad \forall i, k \in I, i < k \quad (4)$$

$$\sum_{c \in C} t_{i,c,d} = 1 \quad \forall i \in I, d \in D \quad (5)$$

$$\sum_{d \in D} t_{i,c,d} = 1 \quad \forall i \in I, c \in C \quad (6)$$

$$x_k^r \leq x_i + (1 - a_{ik}^x)L \quad \forall i, k \in I \quad (7)$$

$$x_i + 1 \leq x_k^r + a_{ik}^x L \quad \forall i, k \in I \quad (8)$$

$$y_k^r \leq y_i + (1 - a_{ik}^y)W \quad \forall i, k \in I \quad (9)$$

$$y_i + 1 \leq y_k^r + a_{ik}^y W \quad \forall i, k \in I \quad (10)$$

$$\frac{x_i + x_i^r}{2} - \frac{x_k + x_k^r}{2} \leq \rho_{ik}^x \quad \forall i, k \in I, i < k \quad (11)$$

$$\frac{x_k + x_k^r}{2} - \frac{x_i + x_i^r}{2} \leq \rho_{ik}^x \quad \forall i, k \in I, i < k \quad (12)$$

$$\frac{y_i + y_i^r}{2} - \frac{y_k + y_k^r}{2} \leq \rho_{ik}^y \quad \forall i, k \in I, i < k \quad (13)$$

$$\frac{y_k + y_k^r}{2} - \frac{y_i + y_i^r}{2} \leq \rho_{ik}^y \quad \forall i, k \in I, i < k \quad (14)$$

$$\frac{x_i + x_i^r}{2} - x^o \leq \rho_{ik}^x \quad \forall i, k \in I, i = k \quad (15)$$

$$x^o - \frac{x_i + x_i^r}{2} \leq \rho_{ik}^x \quad \forall i, k \in I, i = k \quad (16)$$

$$\frac{y_i + y_i^r}{2} - y^o \leq \rho_{ik}^y \quad \forall i, k \in I, i = k \quad (17)$$

$$y^o - \frac{y_i + y_i^r}{2} \leq \rho_{ik}^y \quad \forall i, k \in I, i = k \quad (18)$$

$$t_{i,c,d}, a_{ik}^x, a_{ik}^y \in \{0, 1\} \quad \forall i, k \in I, c \in C, d \in D \quad (19)$$

$$0 \leq x_i \leq x_i^r \leq L \quad \forall i \in I \quad (20)$$

$$0 \leq y_i \leq y_i^r \leq W \quad \forall i \in I \quad (21)$$

$$\rho_{ik}^x, \rho_{ik}^y \geq 0 \quad \forall i, k \in I, i \leq k \quad (22)$$

(2) and (3): collectively ensure each item's coordinates reflects the actual item dimensions given the chosen item orientation.

(4): ensures no overlapping of items i and k given that both items are assigned to the same bin.

(5) and (6): ensure that each item's dimensions is aligned to one of the bin's dimensions, effectively allowing for 90° rotation.

(7) and (8): geometrically ensure if item i is to the right (left) of item k , the right (left) side of item k must be less (greater) than the left (right) side of item i .

(9) and (10): geometrically ensure if item i is to the top (bottom) of item k , the upper (bottom) side of item k must be less (greater) than the bottom (upper) side of item i .

(11) - (14): define ρ_{ik}^x and ρ_{ik}^y as absolute rectilinear distances between the centroids of items i and k .

(15) - (18): define ρ_{ik}^x and ρ_{ik}^y as absolute rectilinear distances between the centroid of each item i and (x^o, y^o) for the diagonal entries π_{ii} .

(19) - (22): establish decision variable domains.

Solution Techniques

Tackling Complexity

Challenge: The 2D-BPP is NP-hard, and the rich objective function makes finding optimal solutions for larger instances computationally intractable.

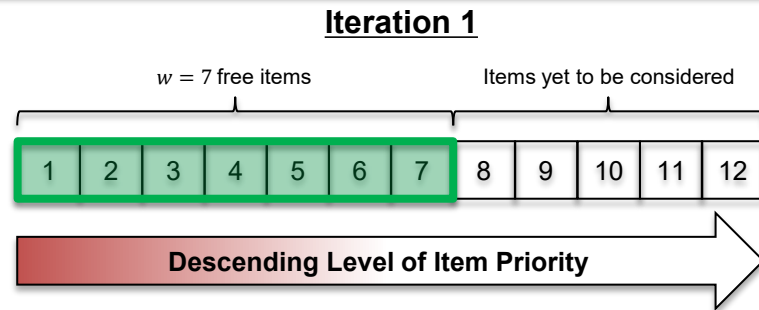
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- Formulate the entire problem as one Mixed-Integer Linear Program.
- Solve using a commercial solver (e.g., Gurobi).
- **Pros:** Can prove global optimality (if given enough time).
- **Cons:** Computation time grows exponentially; intractable for many realistic instances ($> \sim 7$ items within reasonable time).



2. Sliding Window Decomposition:

- **Concept:** Solve a sequence of smaller, overlapping subproblems.
 1. Order all items by a global priority list.
 2. Optimize placement for a “window” of w items.
 3. Fix the highest-priority item(s) in the window.
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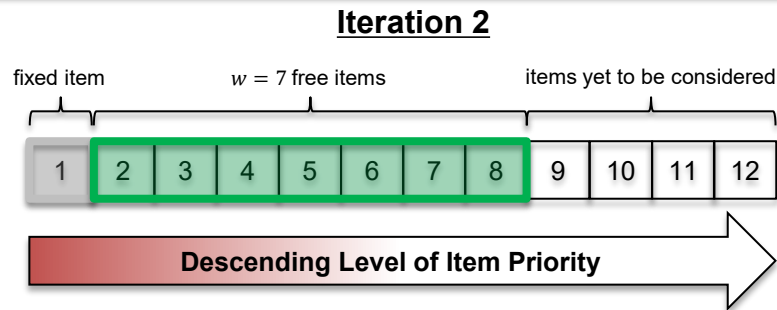
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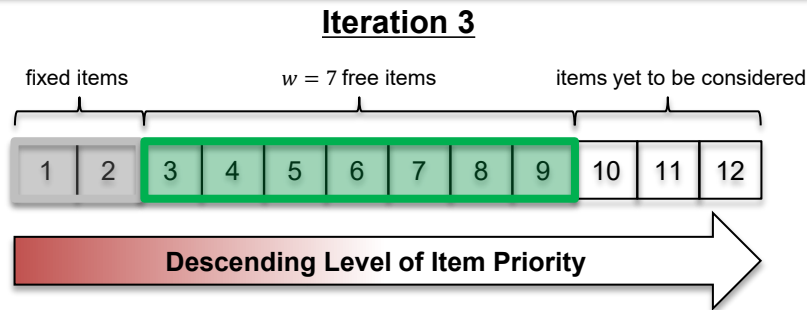
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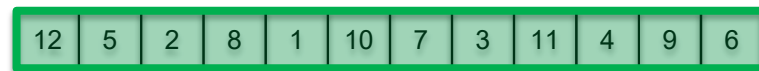
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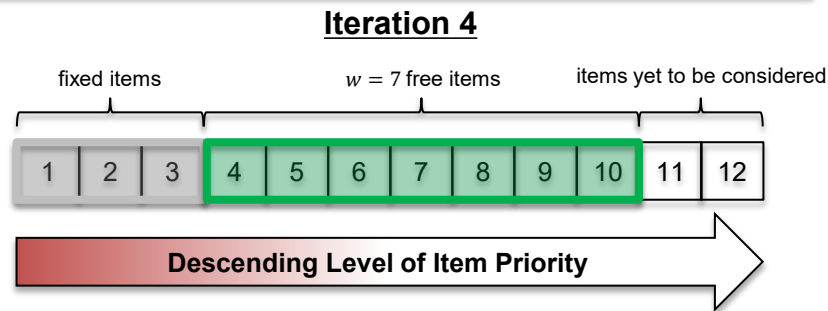
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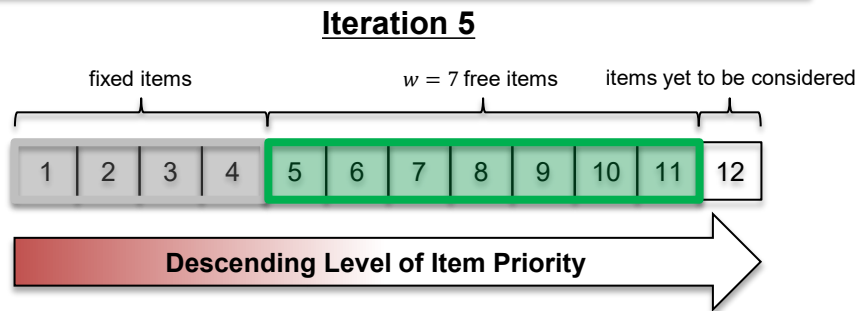
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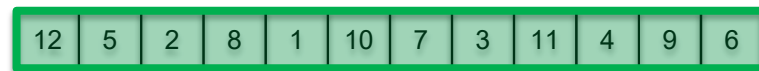
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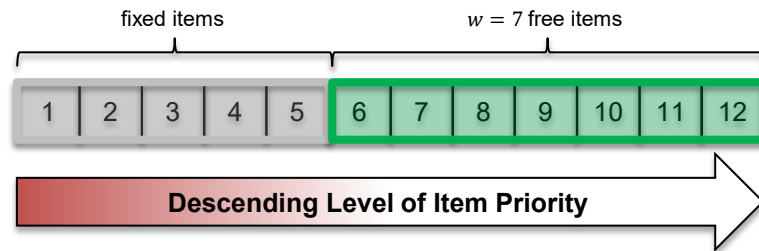
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Iteration 6 (Final)



Lower Bound Methods for Quality Assessment

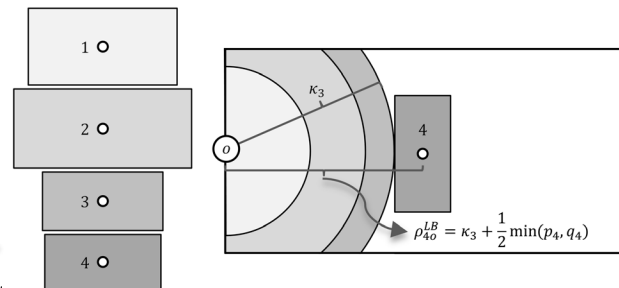
Purpose: Assess solution quality, especially when MILP optimality gaps are large or unavailable for large instances.

Method 1: Grid-Based Lower Bound (GB-LB).

- **Concept:** Deterministic, geometric construction using Euclidean distances ($\leq$ rectilinear distance).
- Iterative “places” items (discretized into cells) greedily based on access priority (π_{ii}).
- Calculates guarantee minimum distances to access point and between items.
- **Theorem 1:** Resulting $LB \leq OPT$. Often provides tighter bound than solver on large instances.

$$LB = \sum_{i=1}^n \pi_{io} \rho_{io}^{LB} + \sum_{i < k} \pi_{ik} \rho_{ik}^{LB}$$

- **Key Idea for ρ_{io}^{LB} (Item-to-Access Point):** 
 - Items form exclusion zones; next item placed at minimum distance considering these zones.
- Provides a geometrically derived benchmark. Often tighter than solver's reported bound on large instances.



Grid-Based Lower Bound (GB-LB) item-to-access-point distance calculation.

Left: Scaled original shapes and centroids of example items 1-4.

Right: Conceptual radial packing of items 1-3 (lighter grays) from access point o within a discretized bin, forming exclusion radius κ_3 . Item 4 (darker gray) is shown pre-discretization to visualize how its lower bound distance ρ_{4o}^{LB} is determined using κ_3 and its minimum dimension.

Method 2: Statistical Lower Bound (Extreme Value Theory).

Experimental Setup

Instance Generation

Objective: Create a comprehensive testbed to rigorously evaluate framework performance and scalability across realistic operational scenarios.

Instance Generation (90 Unique Scenarios):

- **Groups:** 1-5 per instance; Sizes varied (Small: 1-4, Medium: 5-8, Large: 9-12 items).
- **Item Mix:** Varied intra-group size homogeneity (Homogeneous, Weakly/Strongly Heterogeneous); Rectangular items, aspect ratios 1:1 to 1:3.
- **Bin Design:** Single bin with extended length for priority placement flexibility; Center-Left access point for reported results.

Overall Design Aim: To generate a rich dataset that effectively tests the interplay between item/group priorities and geometric packing constraints, ensuring robust evaluation against diverse real-world complexities.

Table 3: Descriptive Statistics for Instance Categories Based on Total Items

Category	# of Groups					Total # of Items				
	Min	Max	Mean	Median	Std	Min	Max	Mean	Median	Std
C1	1	4	1.83	2	0.75	1	16	9.53	10	4.18
C2	2	5	3.43	3.5	0.90	17	26	21.30	21	2.65
C3	3	5	4.47	5	0.63	27	52	34.80	34.5	6.21

Key Findings

1. Group Cohesion Enhances Layout

Research Question: What is the quantitative impact of including group cohesion weights ($\pi_{ik}, i < k$) in the objective?

Analysis: Compared solutions with and without group cohesion terms across 90 instances.

Finding: Group cohesion significantly improves clustering with minimal impact on access point proximity.

- Ignoring cohesion ($\pi_{ik} = 0, i < k$) increases total objective by ~3% (due to item separation penalties).
- Item-to-item distance component inflates by >16% when cohesion is ignored.**
- Negligible impact on access-point proximity term when including group cohesion:** Average improvement of 0.13% (median 0.00%) when cohesion is included.

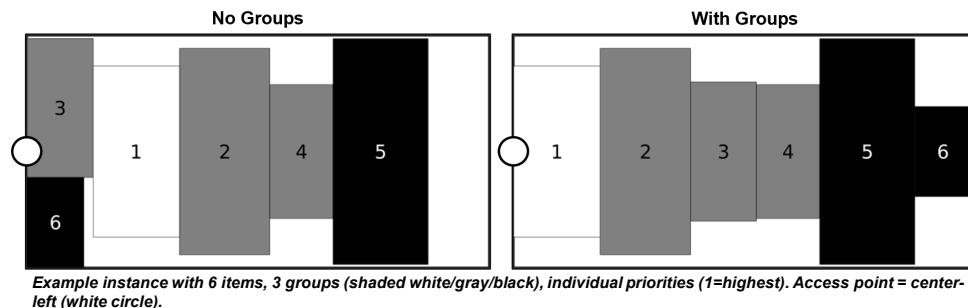


Table 4: Objective Value Percentage Differences From No-Group to Group Consideration Across All 90 Problem Instances.

Objective Component	Offload	mean	median	min	max	std
Full Objective	BL	2.76%	2.40%	-5.75%	9.95%	2.88%
	CL	2.19%	1.94%	-16.40%	13.69%	4.48%
Access Point Proximity Only	BL	0.07%	-0.22%	-11.73%	6.99%	2.83%
	CL	0.13%	0.00%	-22.95%	10.91%	4.59%
Group Cohesion Only	BL	25.27%	23.13%	-12.87%	62.52%	17.45%
	CL	16.33%	13.96%	-15.21%	51.26%	14.88%



Takeaway: *Embedding multi-level priorities (cohesion + access) leads to operationally superior layouts.*

Key Findings

2. Decomposition Yields High-Quality Solutions Quickly

Focus: Efficiency of the Sliding Window Decomposition.

Finding: High-quality solutions are achieved rapidly with appropriate parameters.

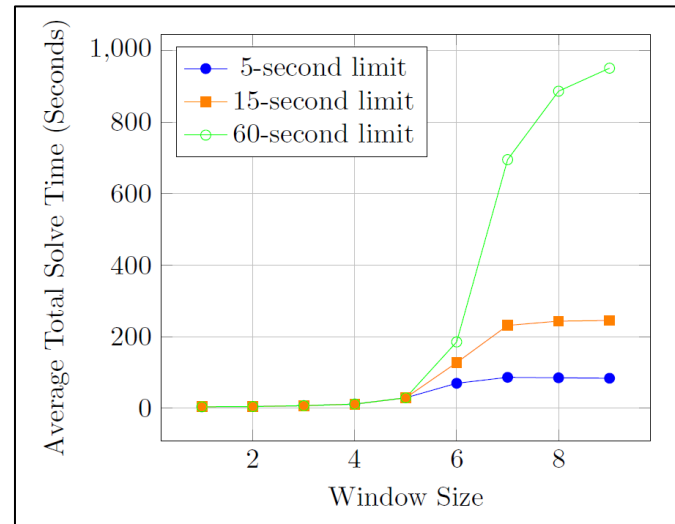
- **Window Size (w) & Time Limit:**
- Small windows ($w = 5$ to 7) with short time limits (5 to 15 seconds per window) achieve solutions often $< 2\%$ deviation from best known.
- Total runtimes can be in seconds or minutes, versus hours for Single MILP to approach similar quality.

Summary statistics for time (in seconds) to prove optimality across all 7-9 item groups treated as standalone instances.

Group Size	Mean	Std	Min	Median	Max
7 Items	9.72	3.88	4.26	8.38	17.32
8 Items	94.56	39.91	30.55	88.51	184.75
9 Items	1279.50	613.31	352.67	1281.83	2853.02



Takeaway: *The sliding window technique is a practically effective and computationally efficient method for realistic problem sizes.*



Average total time vs. window size for windowed solve by time limit.

Key Findings

3. Lower Bounds Provide Quality Insights

Challenge: MILP optimality gaps from solvers can be very large for difficult instances.

Finding: Custom Lower Bounds offer valuable perspectives on solution quality.

Table 4: Comparison of Best Known Objective and Lower Bounds for 8 Largest Problem Instances. Values for Obj, LB, and GB-LB are scaled by 10^5 (e.g., 11.3 represents $1.13e6$). % Gap columns show $((\text{Obj} - \text{Bound}) / \text{Obj}) \times 100\%$. The “Group Size” codes (s = small, m = medium, l = large) denote the size category of each group, and the “Heterogeneity” codes (h = homogeneous, w = weakly heterogeneous, s = strongly heterogeneous) reflect item dimension variation within groups.

Items	Groups	Group Size	Heterogeneity	Obj	LB	GB-LB	LB Gap	GB-LB Gap
52	5	ml111	swhhw	11.3	2.16	9.37	80.80%	16.76%
45	5	ml1ml	wwsss	8.58	1.55	7.30	81.95%	14.94%
44	5	mm111	wwshh	7.97	1.53	6.48	80.82%	18.70%
42	5	slm11	shsws	7.11	1.38	6.13	80.62%	13.76%
42	4	lm11	hhhw	6.31	1.40	5.06	77.87%	19.75%
41	4	lm11	hwsw	7.23	1.49	5.87	79.36%	18.75%
40	5	ml1mm	sshws	6.31	1.27	5.31	79.94%	15.89%
38	5	lmlsl	hshhw	5.24	1.11	4.41	78.90%	15.84%

Table 5: Wilson’s Statistical Lower Bound Results for the Three Largest Instances. Min, Max, and Golden-Alt CI values are scaled by 10^5 .

Instance	Min	Max	KS (Stat, p -val)	AD (Stat, p -val)	Golden-Alt CI	Opt. Gap
42 items	6.31	6.74	(0.056/0.432)	(0.401/0.847)	(6.15, 6.31)	2.61%
45 items	8.58	9.31	(0.034/0.189)	(0.117/0.999)	(8.24, 8.58)	4.58%
52 items	11.3	12.0	(0.076/0.644)	(0.534/0.712)	(10.5, 11.3)	7.62%

Grid-Based LB (GB-LB): Often provides a *tighter* deterministic bound than the solver, especially for large instances. Demonstrates utility of problem-specific geometric bounds.

Statistical LB (Extreme Value Theory): Suggests best-found solutions are closer to the true optimum than deterministic bounds might imply.

Takeaway: *A combination of bounds (solver, GB-LB, statistical) gives a more comprehensive understanding of solution quality, crucial for practical application.*

Key Findings

4. Decomposition Outperforms Single MILP

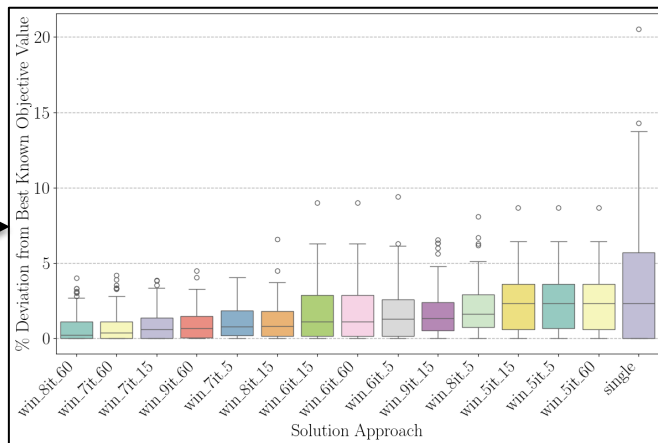
Comparison: Single MILP vs. Sliding Window Decomposition across 90 diverse instances.

Metrics: Solution quality (% deviation from best known) & computation time.

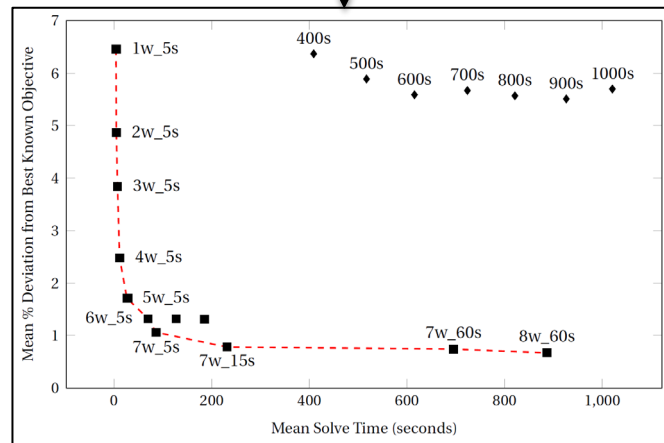
Finding: Decomposition methods dominate the Single MILP.

Tighter Distributions: Decomposition methods show much less variance in solution quality.

- Pareto Efficiency:** Sliding window solutions consistently form the Pareto frontier.



Boxplot of % deviation from best known objective, ordered by increasing median.



Pareto curve showing the relationship between mean solution time and mean % deviation from the best obtained objective values across all instances.

- Black squares = sliding window decomposition solutions.
- Black diamonds = single MILP solutions recorded at various time intervals.
- Dashed line indicates the Pareto-efficient frontier.

Takeaway: *Sliding window decomposition methods achieve better solutions in far less time than a single MILP using a commercial solver.*

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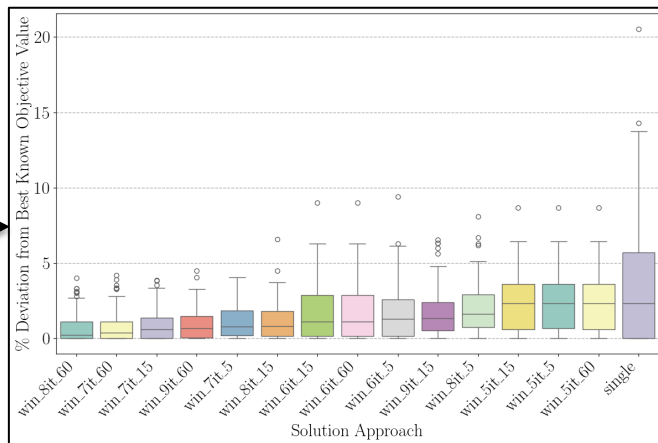
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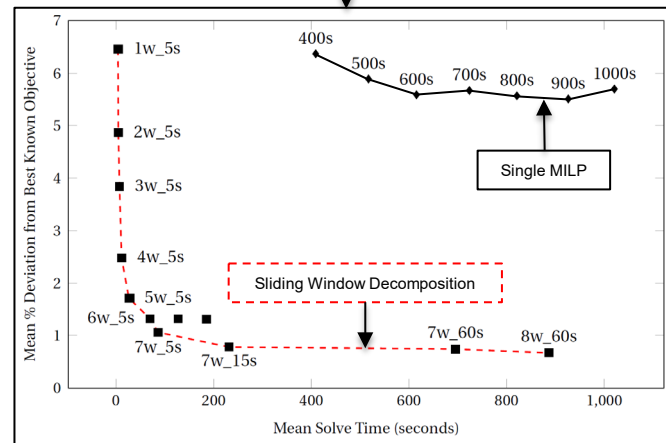
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Conclusion

Contributions & Future Work

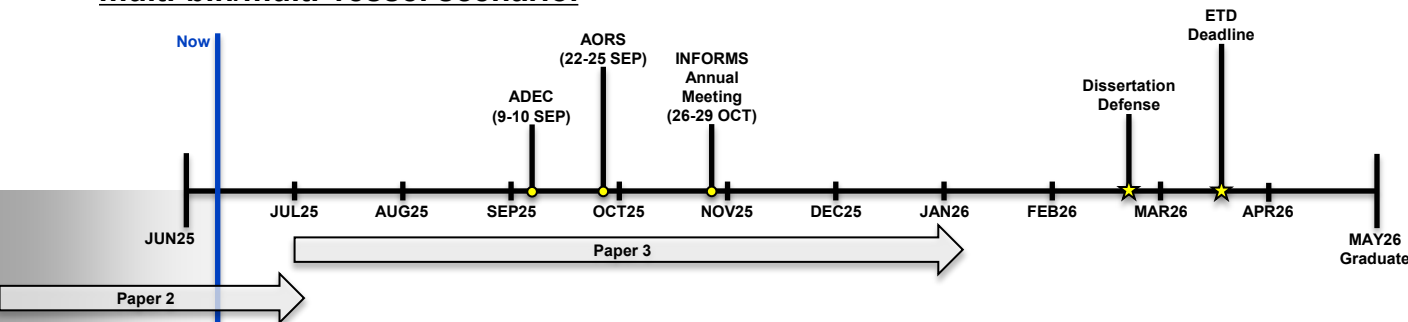
Key Contributions:

- Introduced a **novel 2D-BPP framework** that uniquely integrates multi-level spatial priorities (access point & group cohesion) via a flexible prioritization matrix.
- Demonstrated an **effective sliding window decomposition** that rapidly yields high-quality, priority-driven layouts, significantly outperforming direct MILP solutions.
- Established methods for **rigorous solution quality assessment** using new geometric lower bound and complementary statistical lower bound, enhancing confidence in our solution approaches.

Limitations:

- Primarily focused on a single bin with 2D rectangular items.
- This decomposition method provides practical solutions without global optimality guarantees (though LBs help assess quality).

Future Directions & Ongoing Work: Applying this prioritized framework to complex military combat loading and a multi-bin/multi-vessel scenario.



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